RNN: Recurrent Neural Networks

feat. GPT

csv 파일

7, 6, 42

$7 \times 6 \rightarrow 42$

입력 정답

이전에 어떤 구구단을 말했는지 관계없다.

“

과 →

입력 정답

“

동해물과 →

입력 정답

“

동해물과 → 백

입력 정답

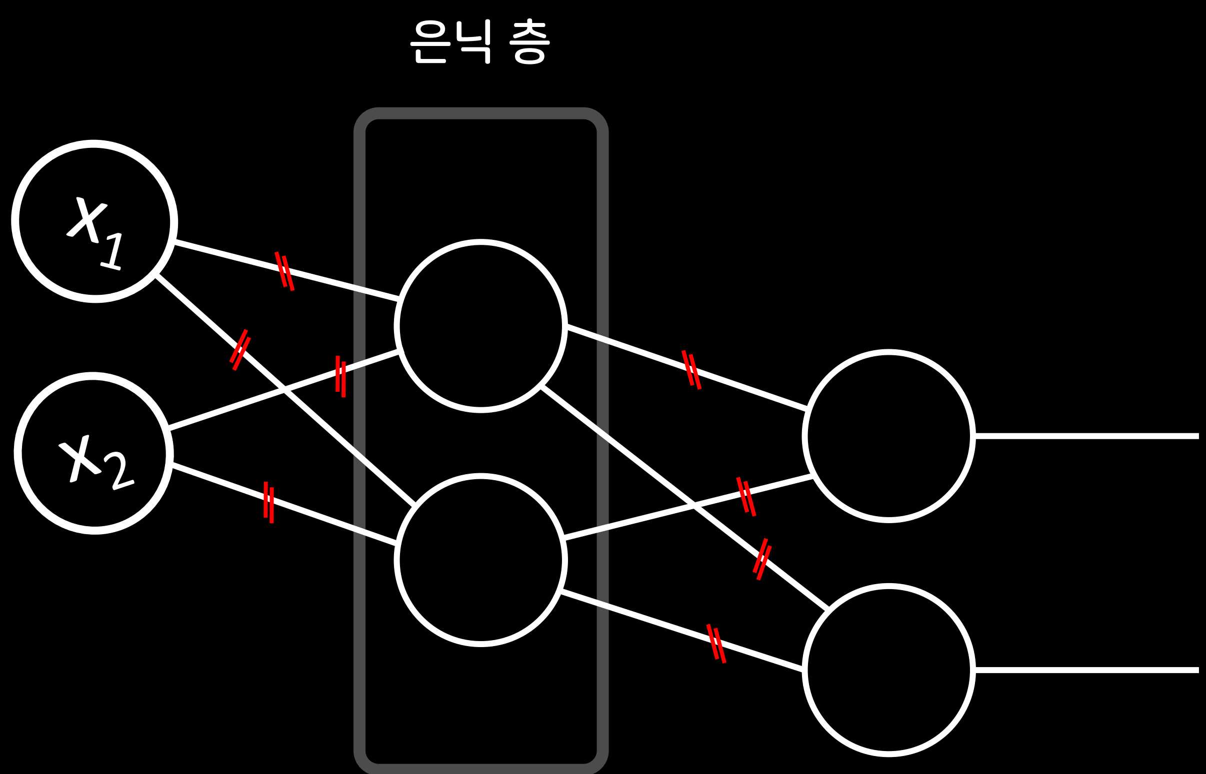
1. 동해 물과 백두산이 마르고 닳도록
하느님이 보우하사 우리나라 만세
무궁화 삼천리 화려 강산
대한 사람 대한으로 길이 보전하세

"무슨 말인지 알겠니?"

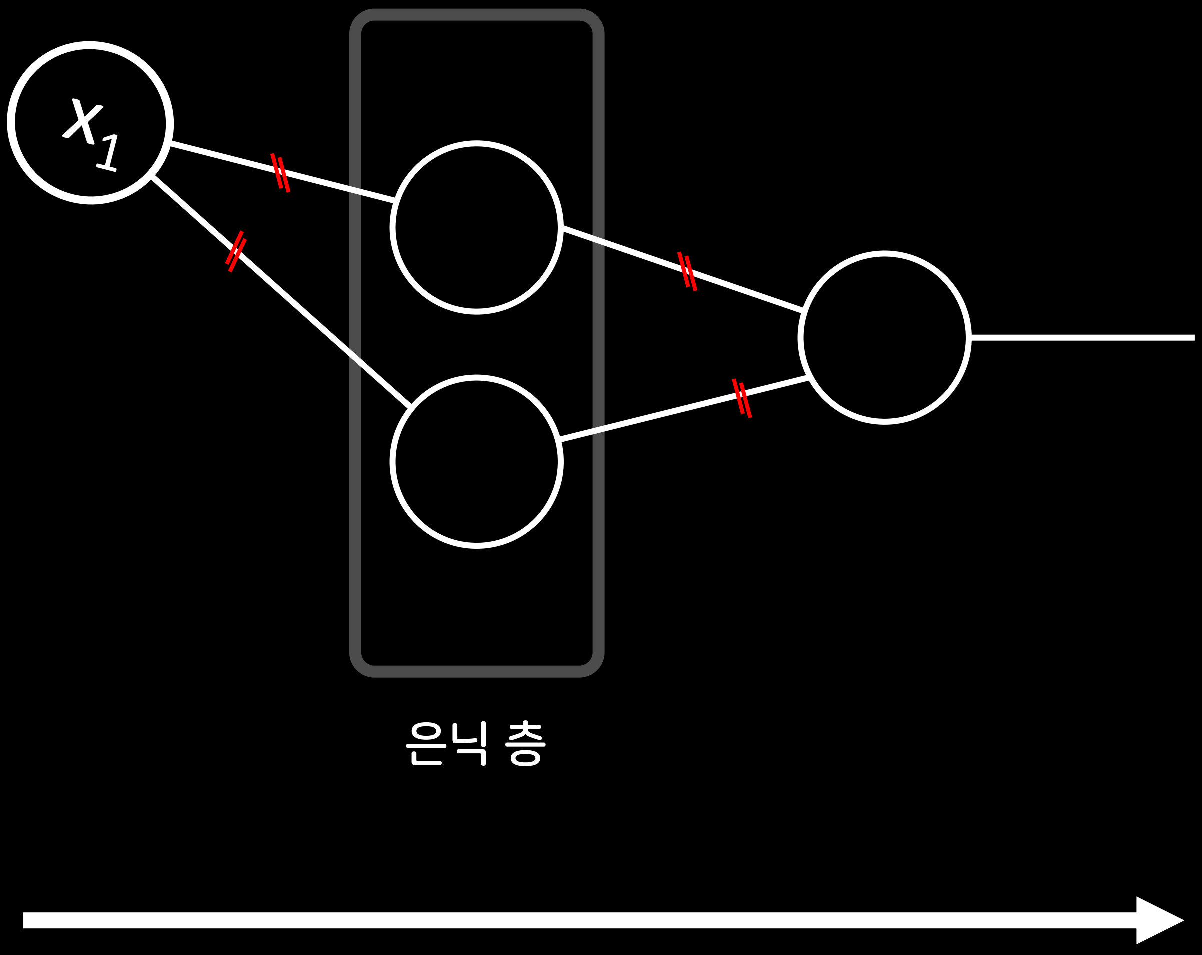
“

이전 내용을 기억하는 법

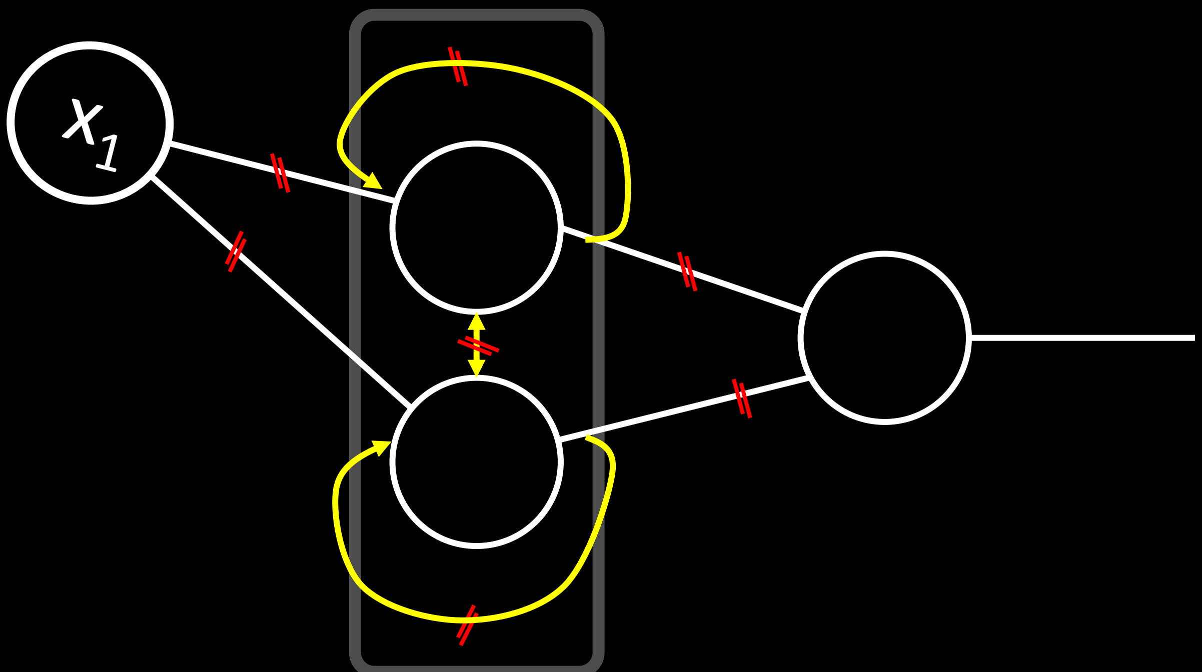
지금(혹은 이전) 상태에 따라
다음에 나올 내용이 달라진다.


은닉 층
x_1
x_2

순방향 신경망 Feedforward Neural Networks

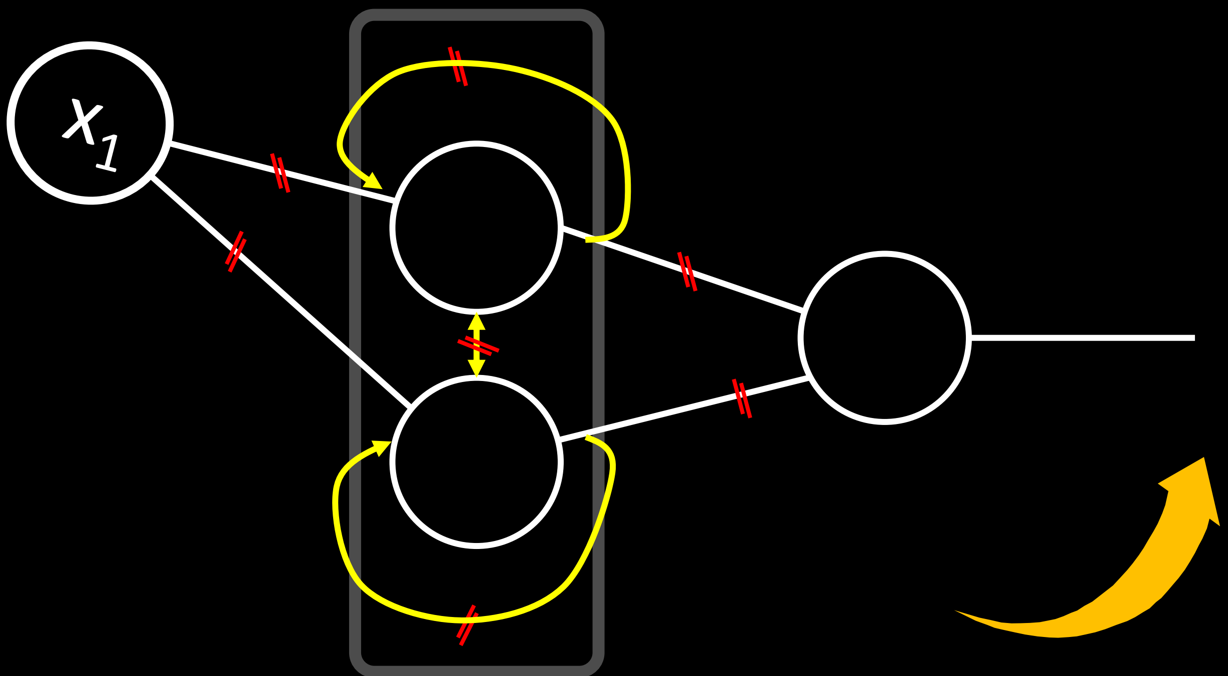


순환 신경망 Recurrent Neural Networks



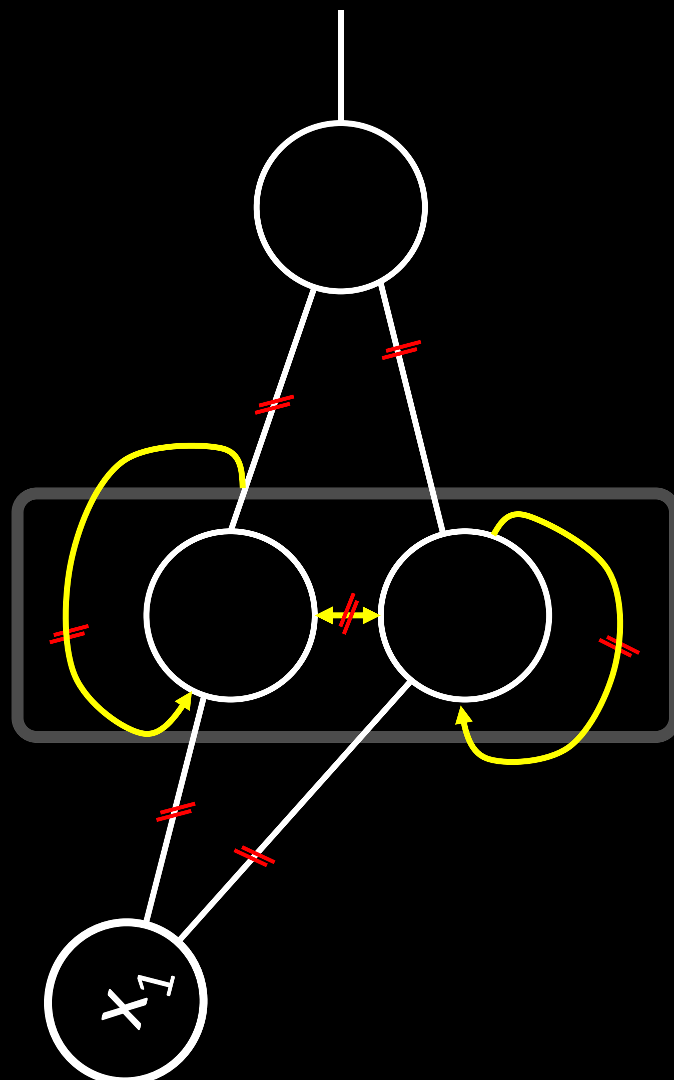
은닉층의
상태정보

순환 신경망 Recurrent Neural Networks



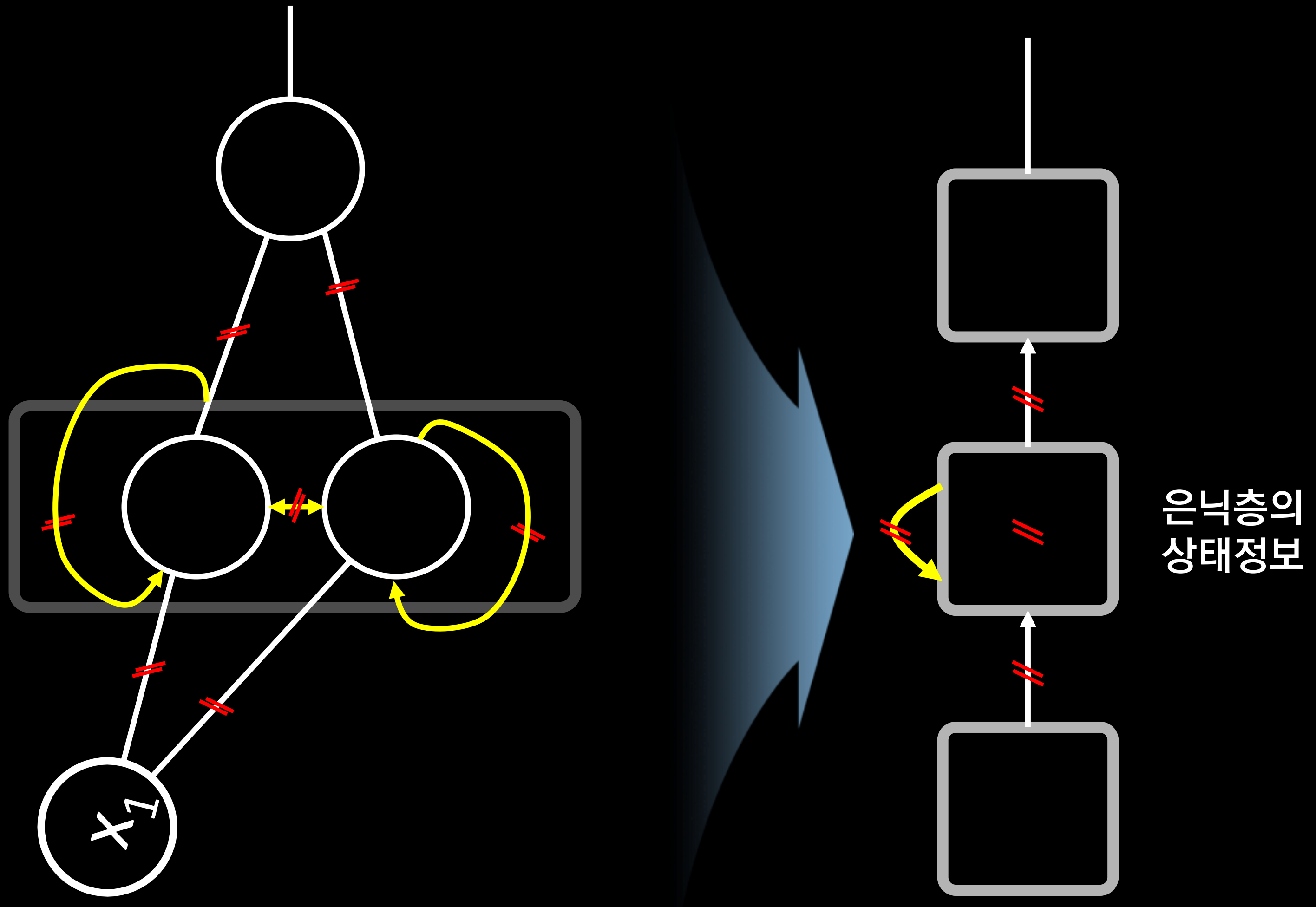
은닉층의
상태정보

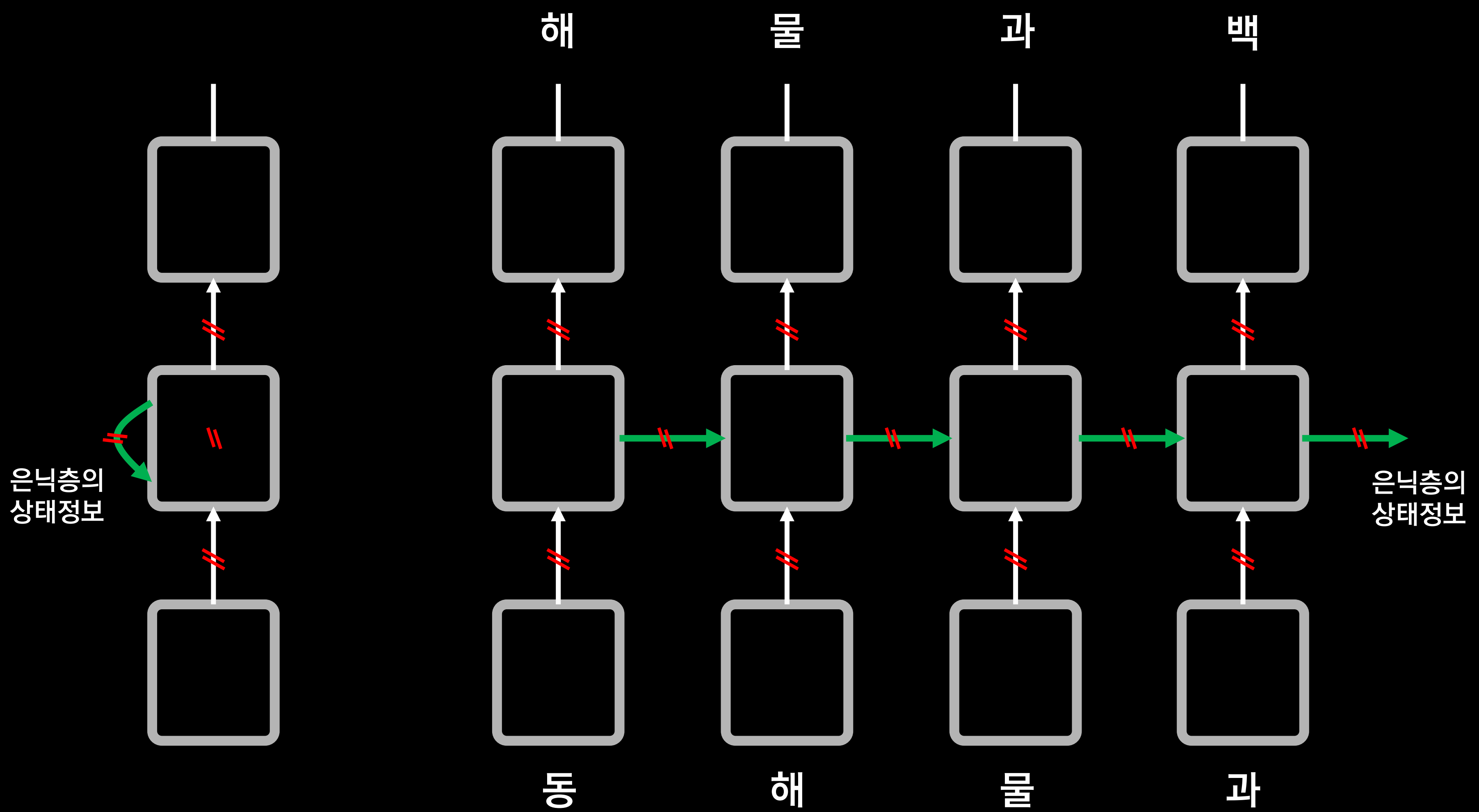
순환 신경망 Recurrent Neural Networks



은닉층의
상태정보

순환 신경망 Recurrent Neural Networks





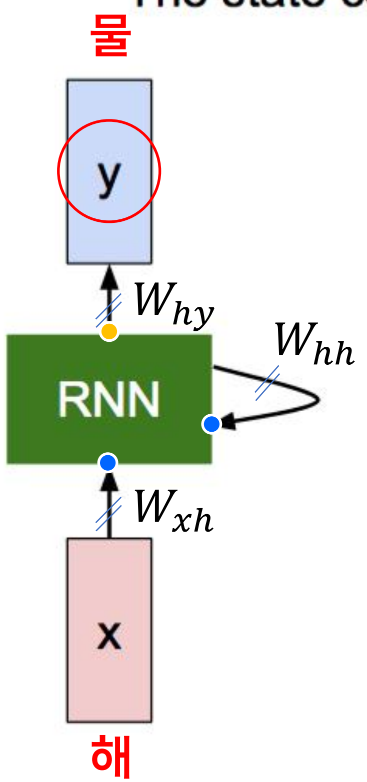
Unfolded

시냅스(파라미터)는 몇 개?

바닐라 아이스크림

(Vanilla) Recurrent Neural Network

The state consists of a single *"hidden"* vector **h**:

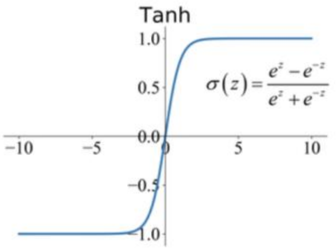


$$h_t = f_W(h_{t-1}, x_t)$$

현재 상태 이전 상태 입력

$$h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t)$$

$$y_t = W_{hy}h_t$$

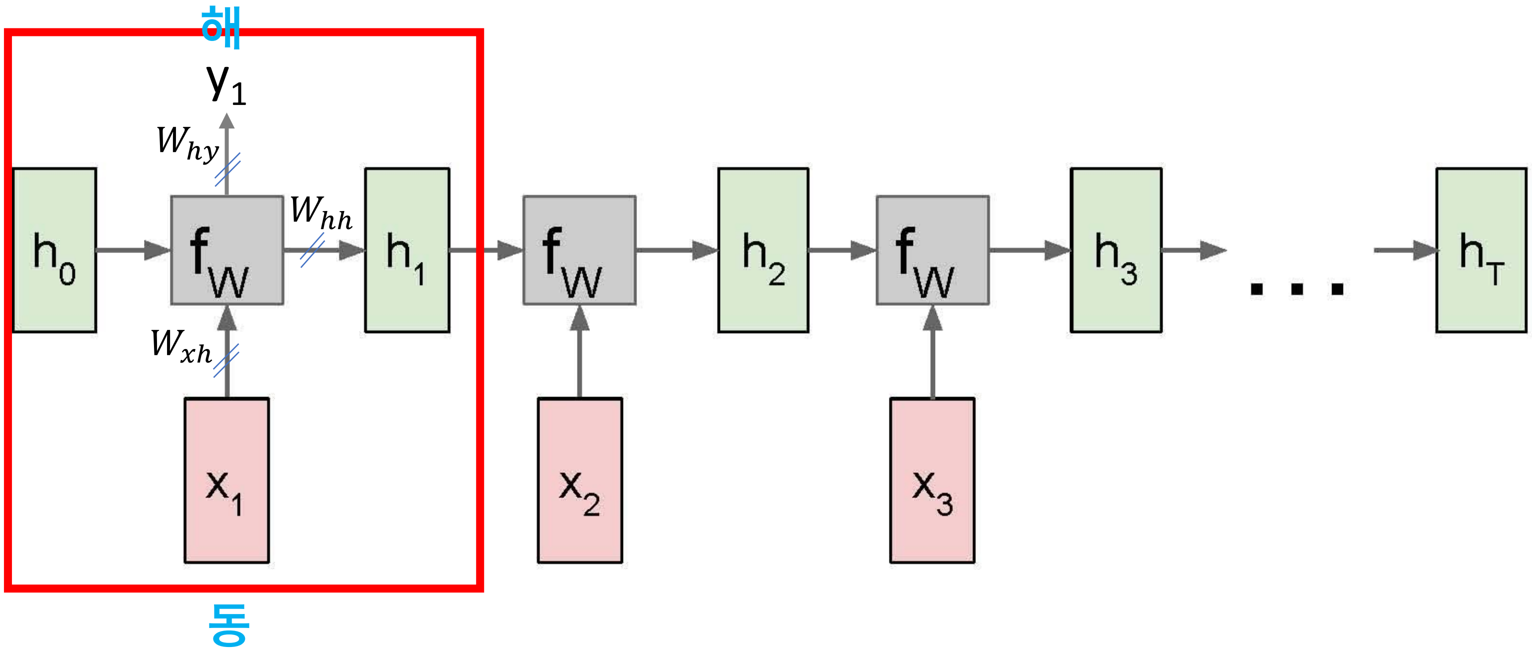


RNN: Computational Graph (Unfolding)

$$h_1 = f_W(h_0, x_1)$$

$$h_1 = tanh(W_{hh}h_0 + W_{xh}x_1)$$

$$y_1 = W_{hy}h_1$$

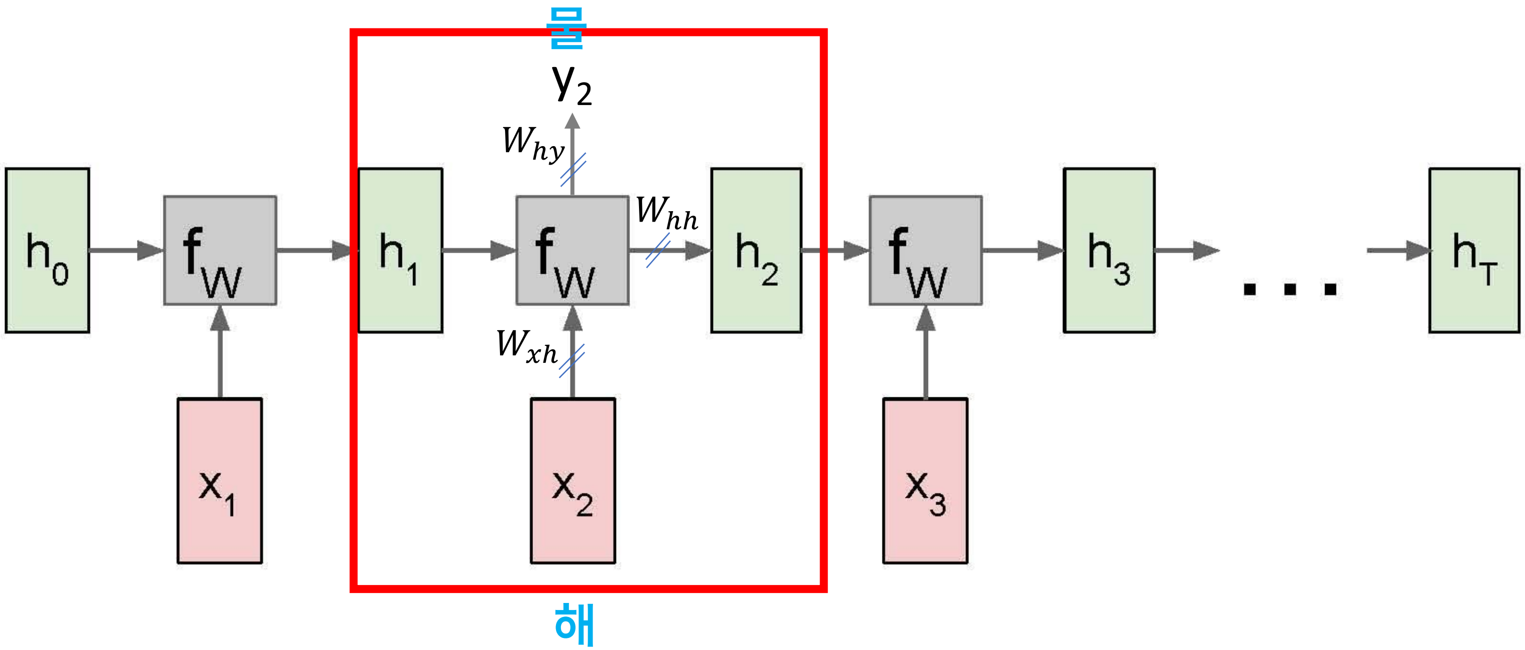


RNN: Computational Graph (Unfolding)

$$h_2 = f_W(h_1, x_2)$$

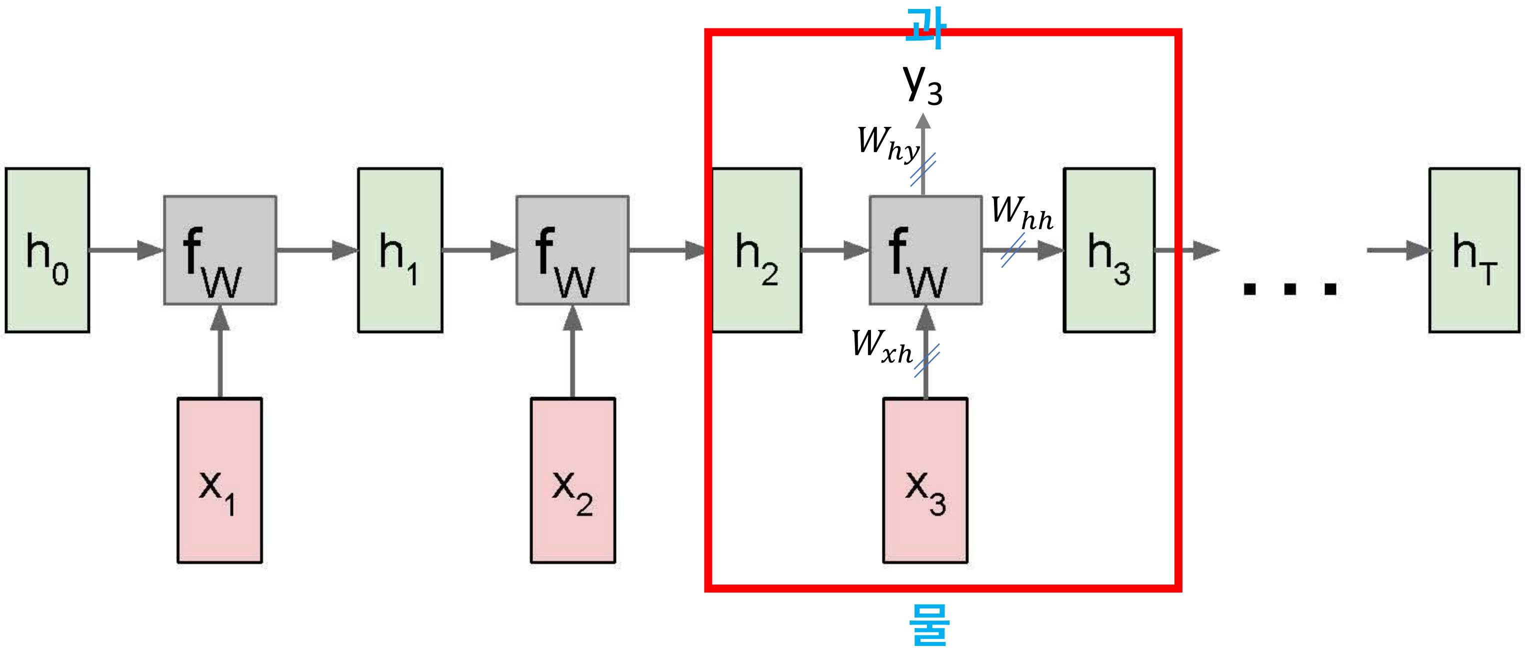
$$h_2 = tanh(W_{hh}h_1 + W_{xh}x_2)$$

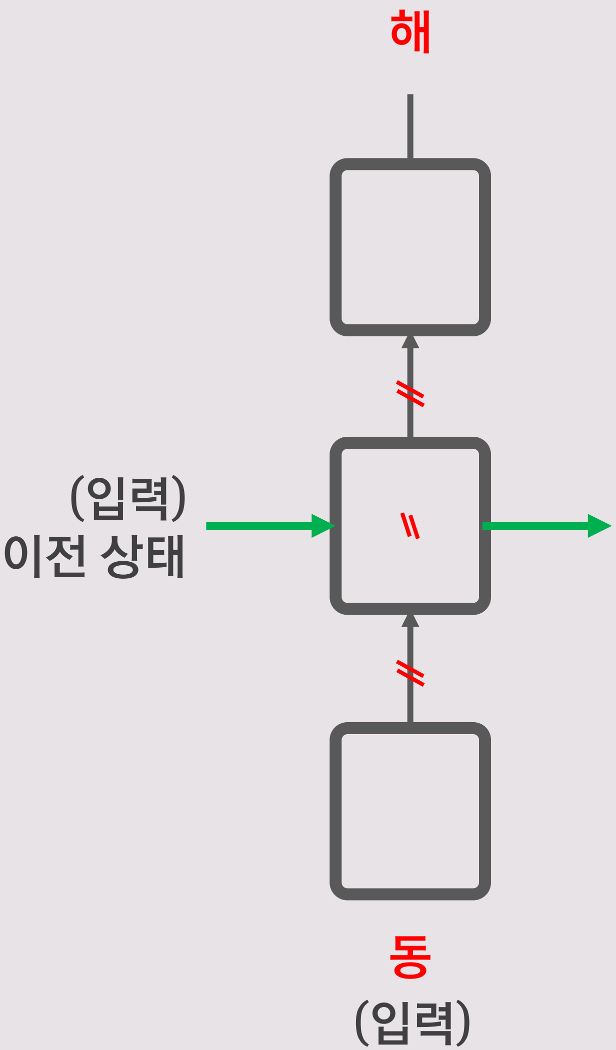
$$y_2 = W_{hy}h_2$$

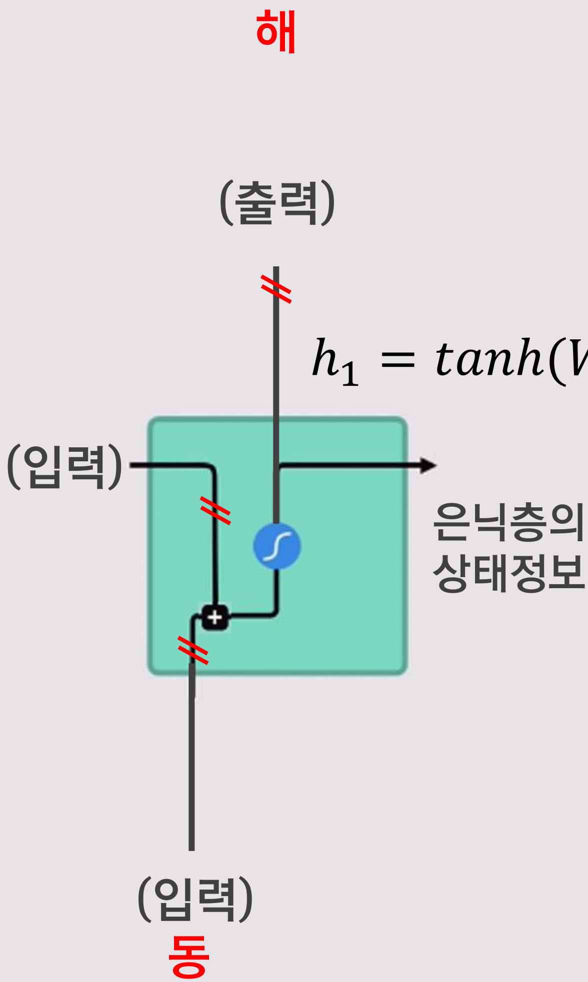


RNN: Computational Graph (Unfolding)

$$h_3 = f_W(h_2, x_3)$$
$$h_3 = tanh(W_{hh}h_2 + W_{xh}x_3)$$
$$y_3 = W_{hy}h_3$$



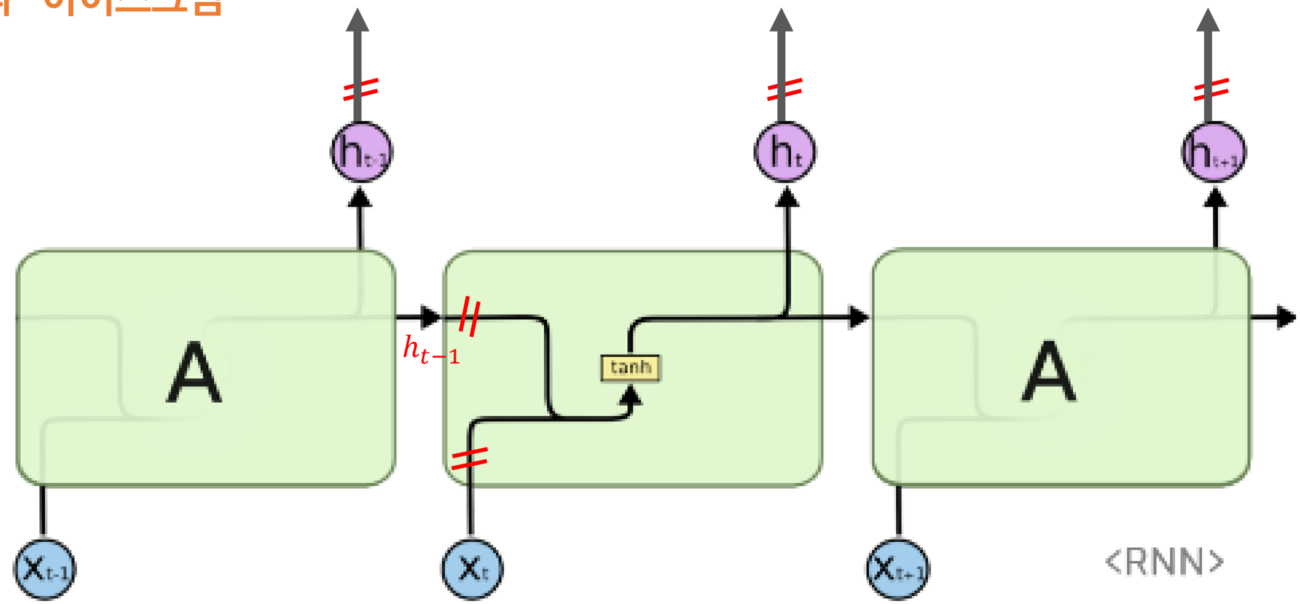




$$h_1 = tanh(W_{hh}h_0 + W_{xh}x_1)$$

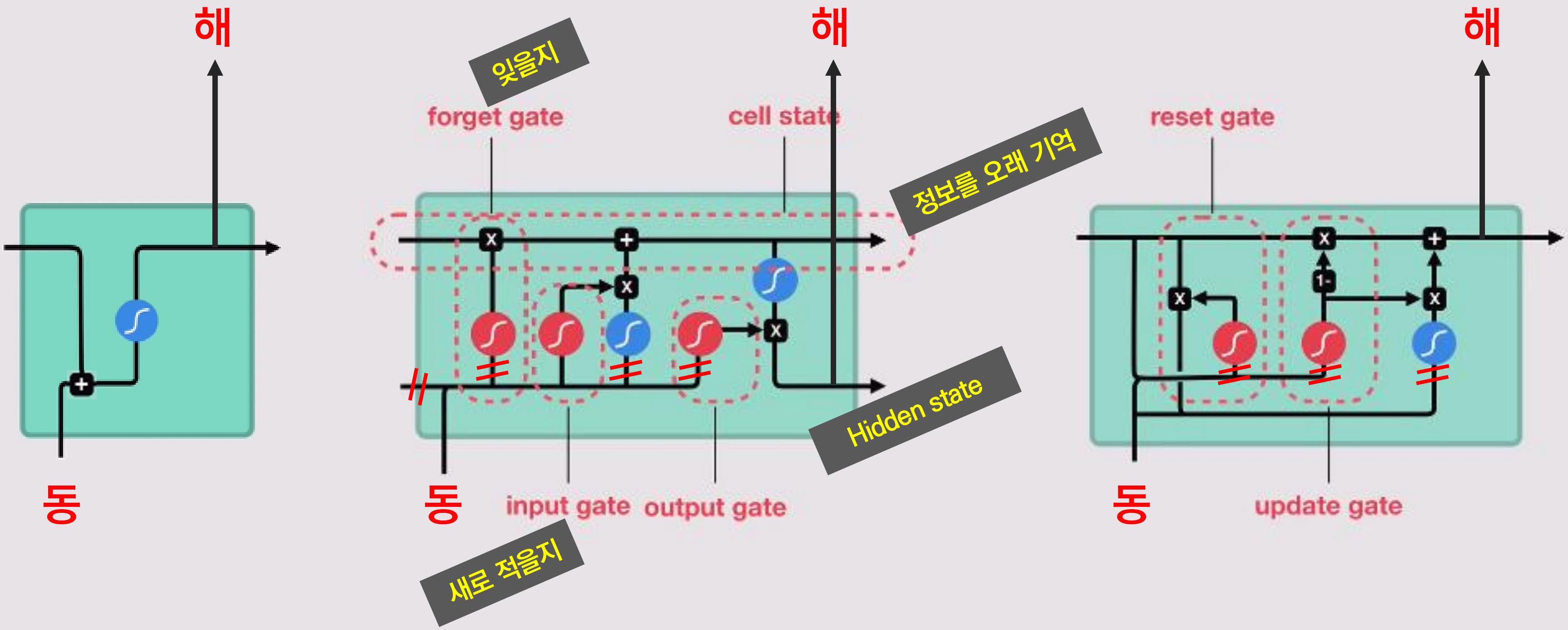
Vanilla RNN

"바닐라" 아이스크림



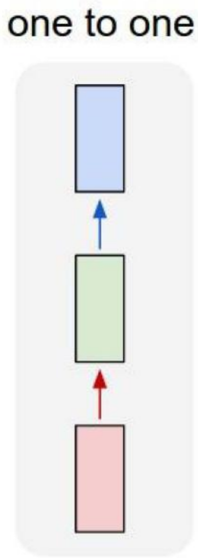
$$h_1 = tanh(W_{hh}h_0 + W_{xh}x_1)$$

Vanilla RNN & LSTM & GRU



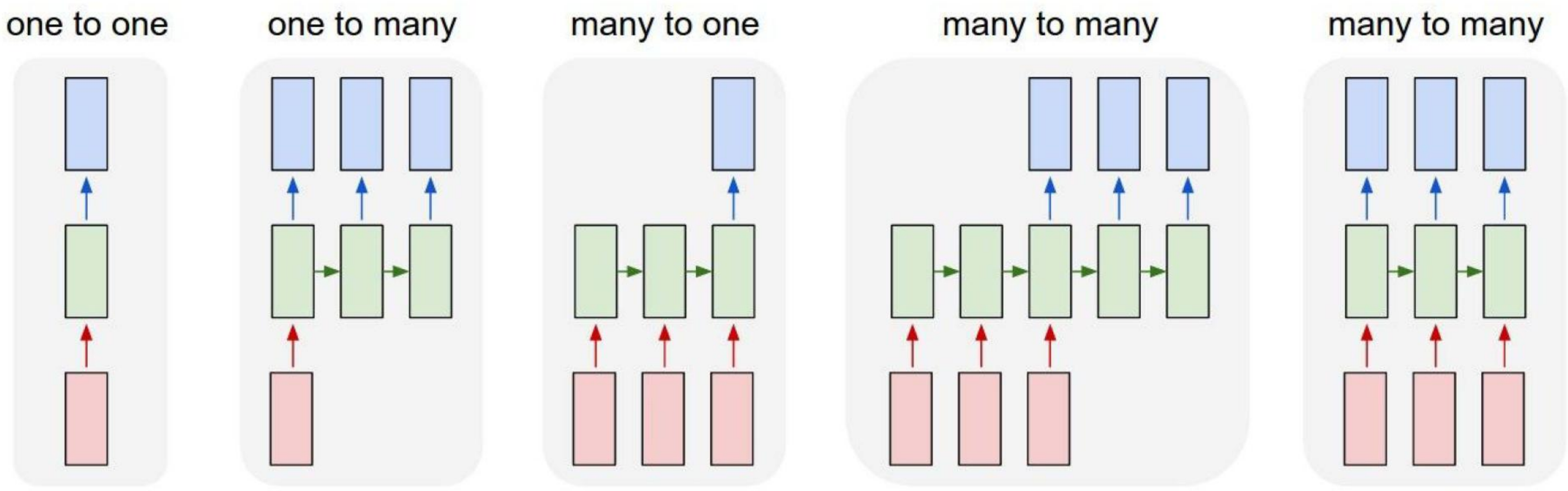
RNN	LSTM	GRU
Short-Term Memory	Long Short-Term Memory	Gated Recurrent Units
이전 내용을 잘 기억 못함	개선되었으나 복잡	간단한 구조로 개선

“Vanilla” Neural Network



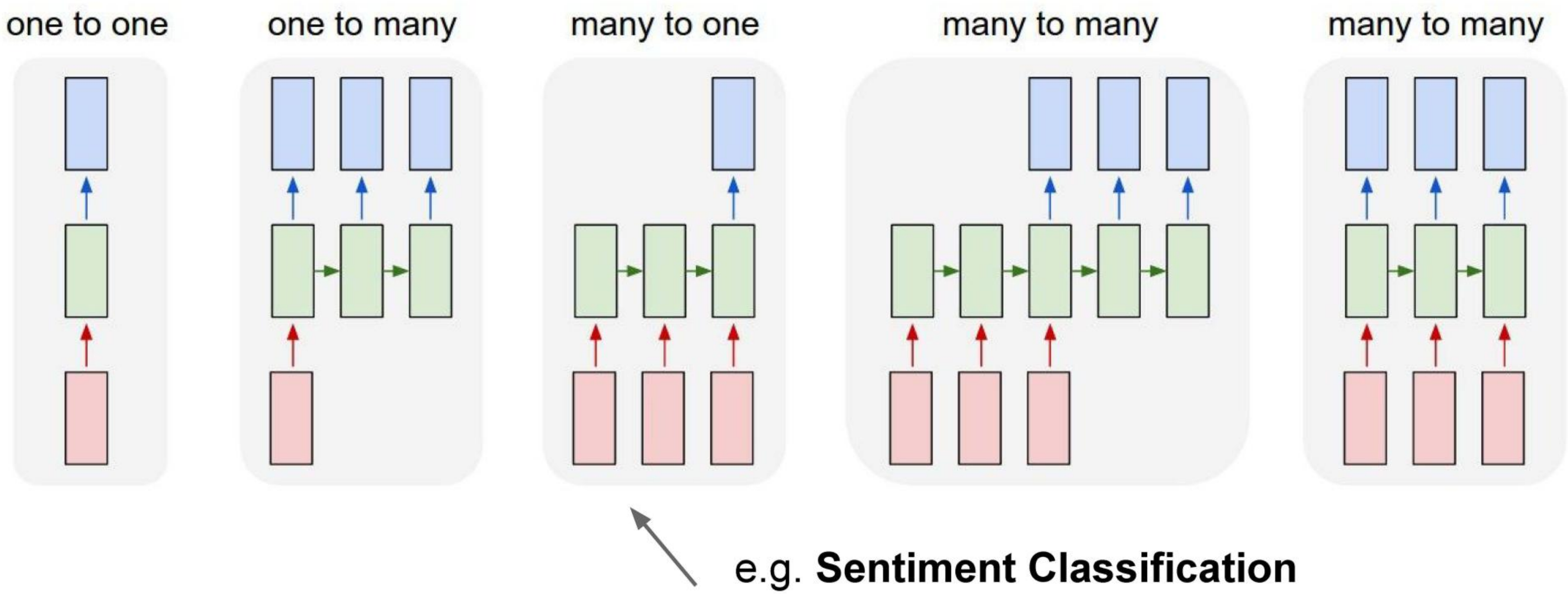
Vanilla Neural Networks

Recurrent Neural Networks: Process Sequences



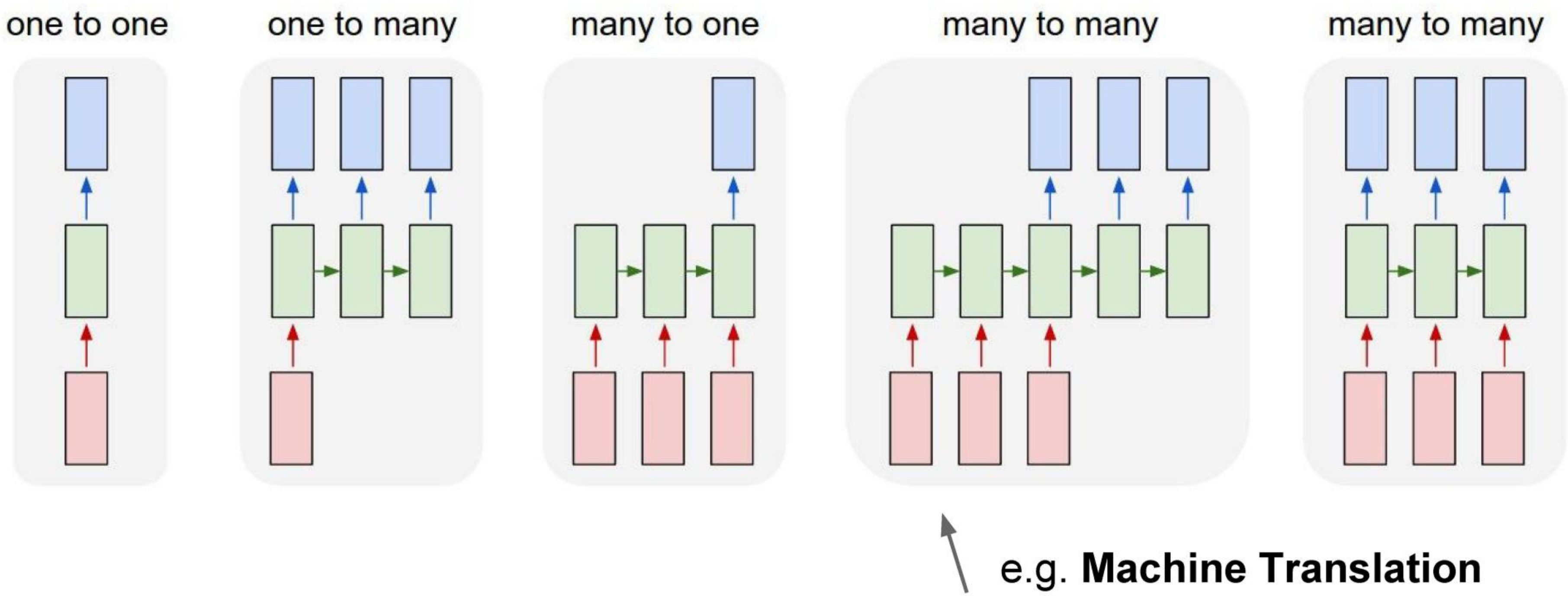
e.g. **Image Captioning**
image -> sequence of words

Recurrent Neural Networks: Process Sequences



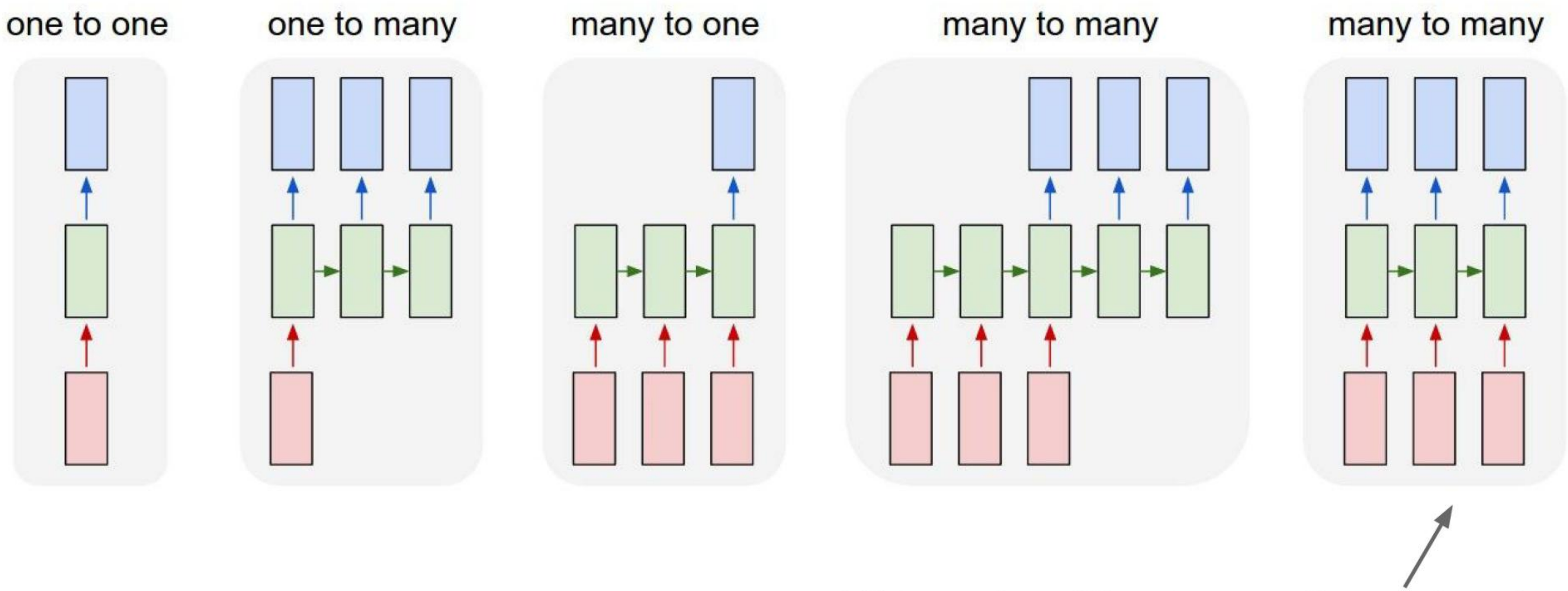
e.g. **Sentiment Classification**
sequence of words -> sentiment

Recurrent Neural Networks: Process Sequences



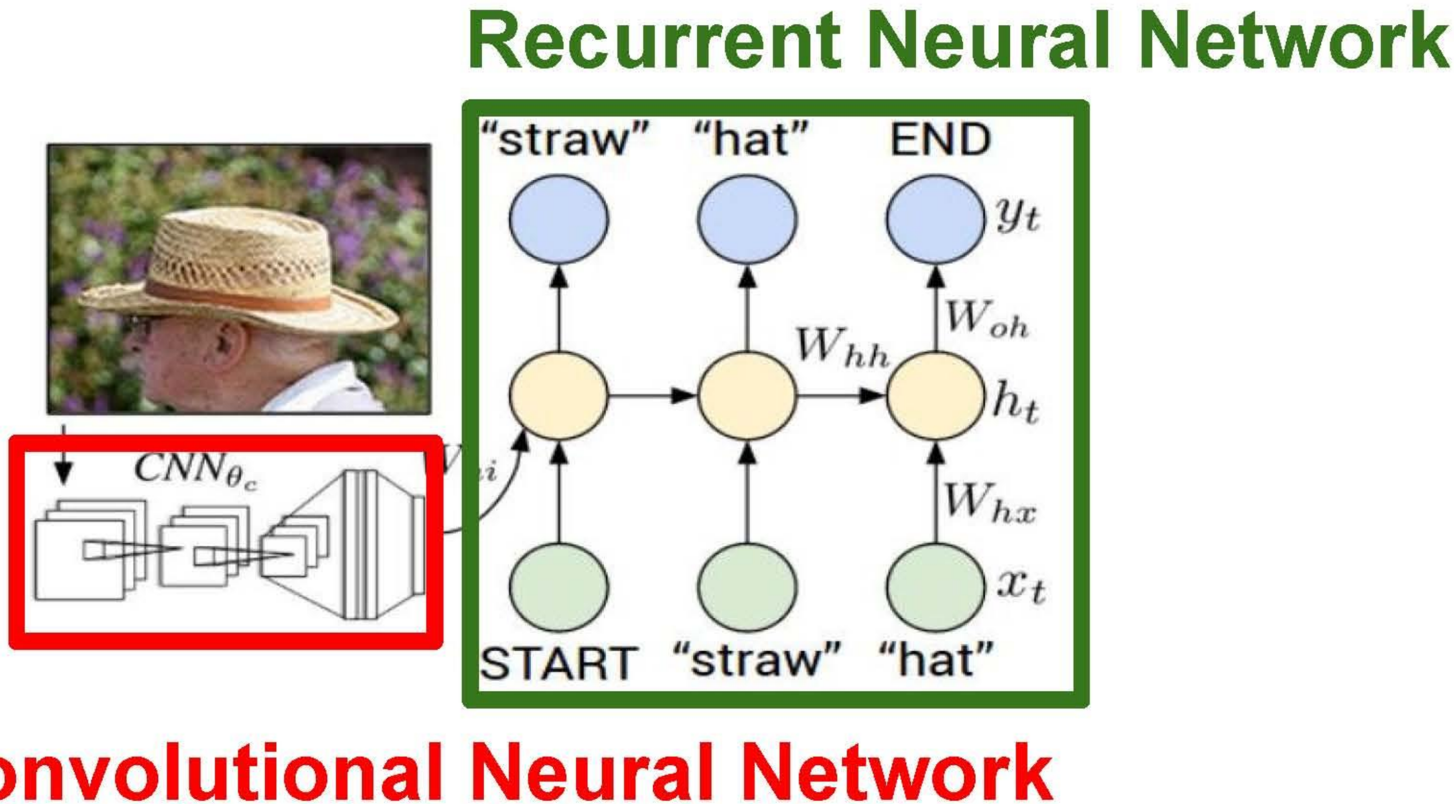
e.g. **Machine Translation**
seq of words -> seq of words

Recurrent Neural Networks: Process Sequences



e.g. **Video classification on frame level**

CNN + RNN


Recurrent Neural Network
"straw" "hat" END
y_t
W_{oh}
W_{hh}
h_t
W_{hx}
x_t
START "straw" "hat"
CNN_{θ_c}
Convolutional Neural Network

CNN + RNN

Image Captioning: Example Results

Captions generated using neuraltalk2
All images are CC0 Public domain:
cat suitcase, cat tree, dog, bear,
surfers, tennis, giraffe, motorcycle

A cat sitting on a suitcase on the floor

A cat is sitting on a tree branch

A dog is running in the grass with a frisbee

A white teddy bear sitting in the grass

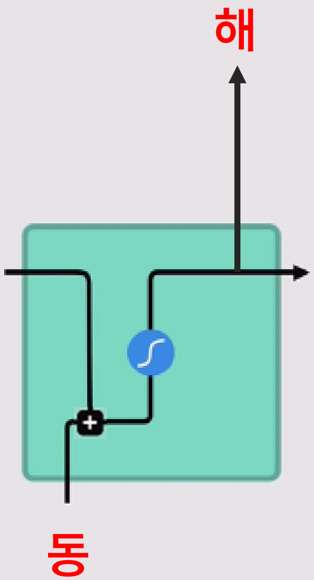
Two people walking on the beach with surfboards

A tennis player in action on the court

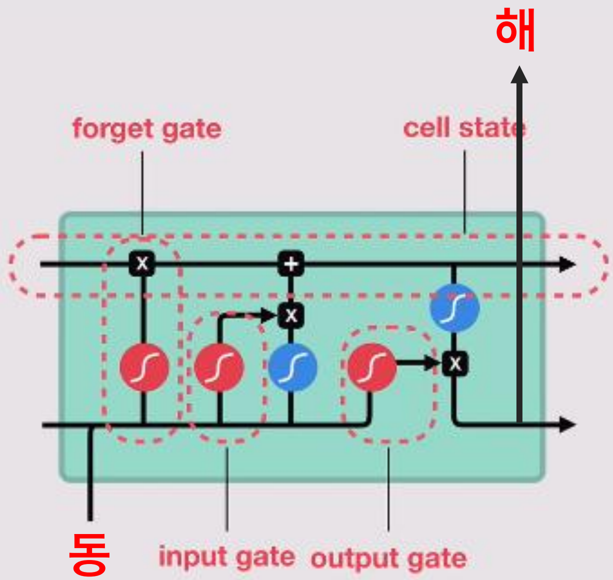
Two giraffes standing in a grassy field

A man riding a dirt bike on a dirt track

Vanilla RNN & LSTM & GRU



RNN
Short-Term Memory
이전 내용을 잘 기억 못함

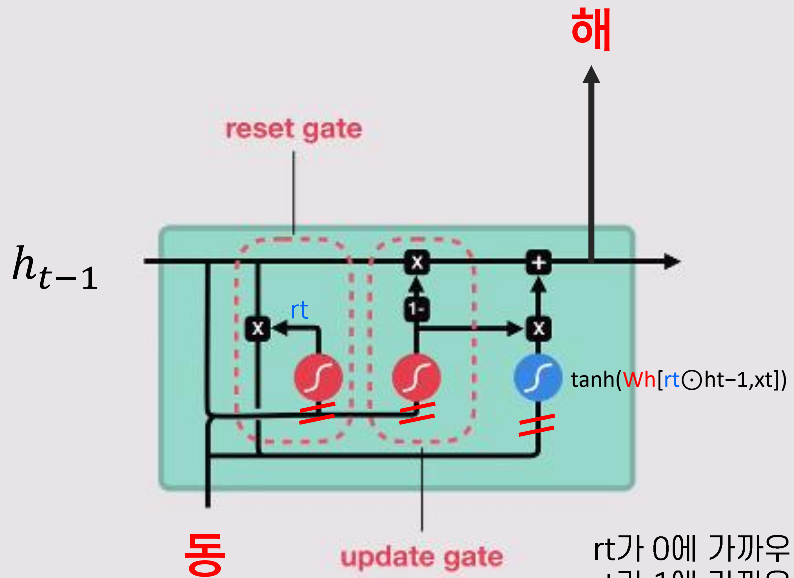


LSTM
Long Short-Term Memory
개선되었으나 복잡

Forget gate:
- 시그모이드는 “중요해서 잊지 말아야 하는지(1에 가까운 값)” 또는 “중요하지 않아서 잊어야 하는지(0에 가까운 값)” 결정
- 결과(0에 가까운 숫자)를 기존 기억에 곱해주면 기억은 0이 되거나 아주 작아져서 사실상 잊혀지는 것

Input gate:
- 빨간 sigmoid: 새로 들어온 정보가 얼마나 중요한지 판단하여 ‘기억’할 지 결정
- 파란 sigmoid: 기억에 추가할 새로운 정보를 만들어 기존 기억에 합칠 준비

Vanilla RNN & LSTM & GRU



나는 어제 피자를 먹었다.
그런데 오늘 날씨는 춥다.

나는 어제 피자를 먹었다.
그것은 정말 맛있었다.

rt가 0에 가까우면 0×ht−1, 이전 기억 reset
rt가 1에 가까우면 1×ht−1, 이전 기억 그대로 기억

⊙: Element-wise multiplication

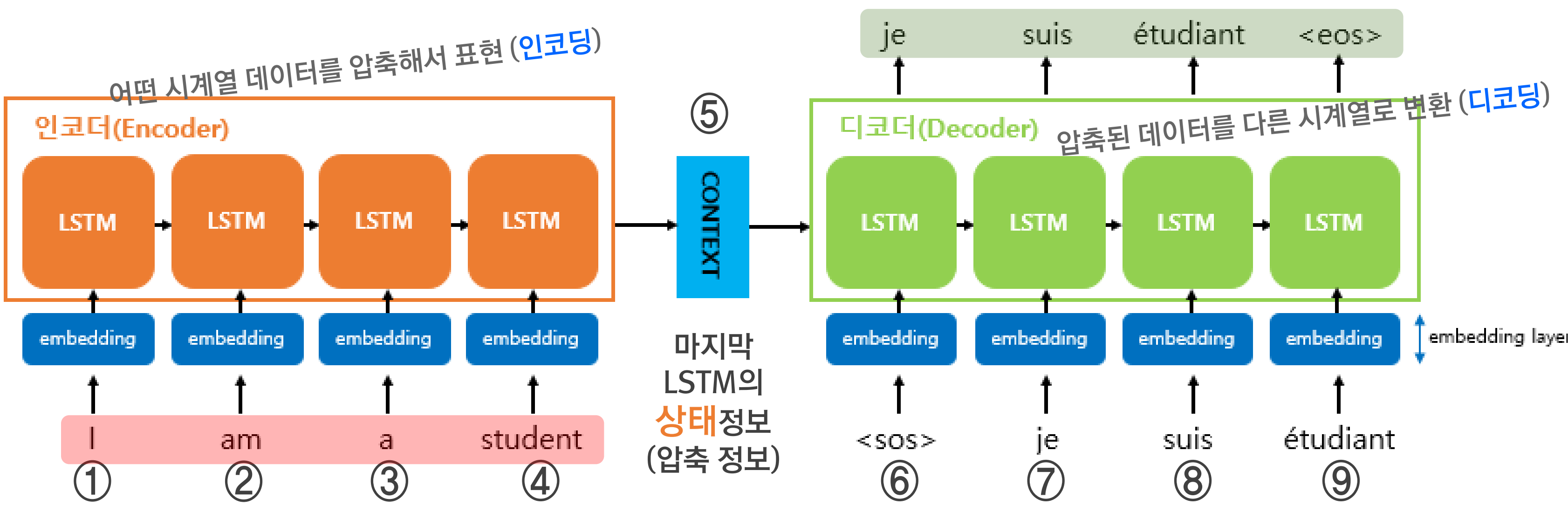
GRU

Gated Recurrent Units

간단한 구조로 개선

Seq2Seq

(영어 입력) I am a student 문장 시퀀스 →
(불어 정답) Je suis etudiant 문장 시퀀스
문장 변환 (Transformation)



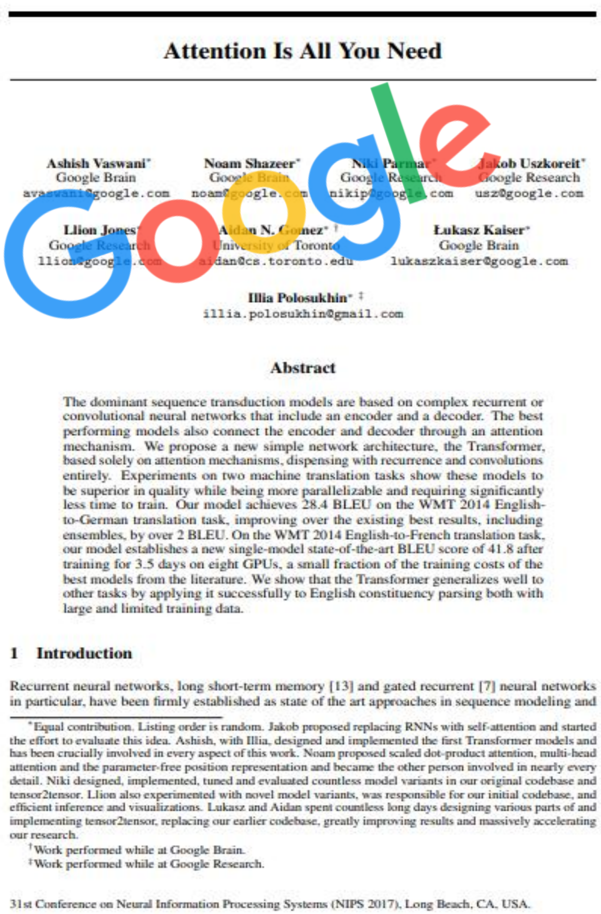
Encoder → Context Vector → Decoder

우리 아이는 빨간색 운동화를 좋아한다.
그래서 그것이 다 닳을 때까지 신었다.

- 순환신경망은 오래 전 내용은 기억을 못함.
- 중요한 것이 무엇인지 모름.
- 순환신경망은 위에서 그것이 우리인지, 아이인지, 운동화인지 알 수 없음.
- 반면 트랜스포머는 **단어끼리 연관성 계산, 의미를 찾아 학습** (Attention, 주의)
- 왕자님과? → 어떤 단어를 들으면 연관된 단어가 생각(이 집중, attention)난다.

트랜스포머

31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA.



Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

1 Introduction

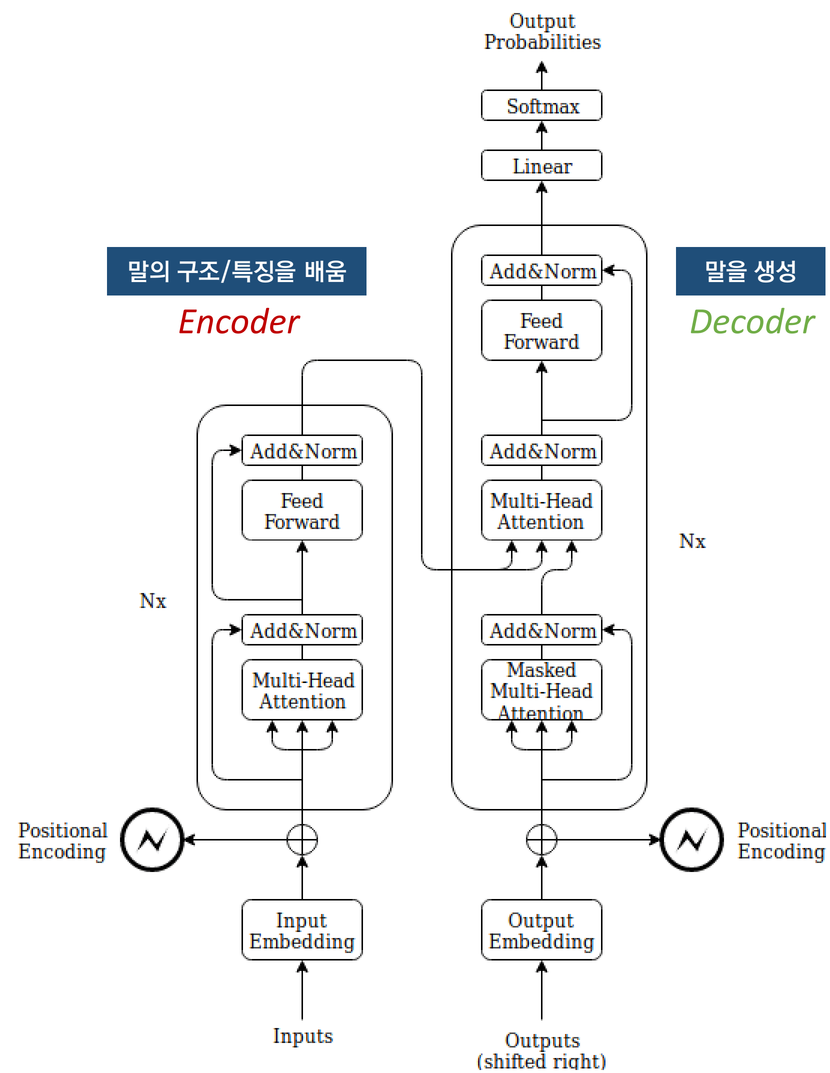
Recurrent neural networks, long short-term memory [13] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and

*Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

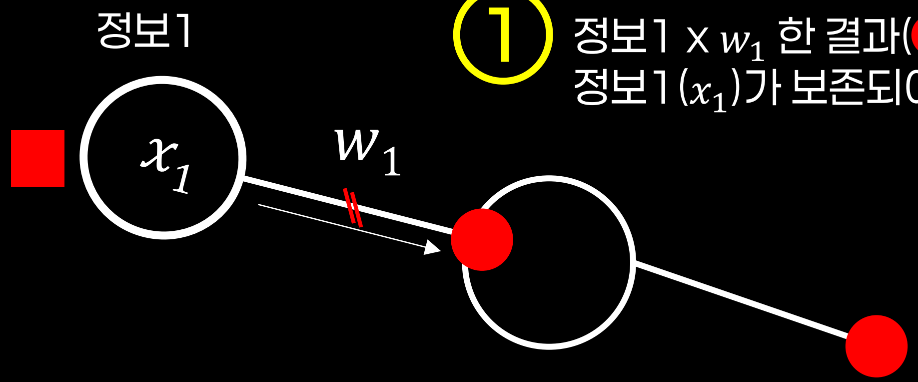
†Work performed while at Google Brain.

‡Work performed while at Google Research.

31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA.



벡터의 3가지 성질 (1/3)

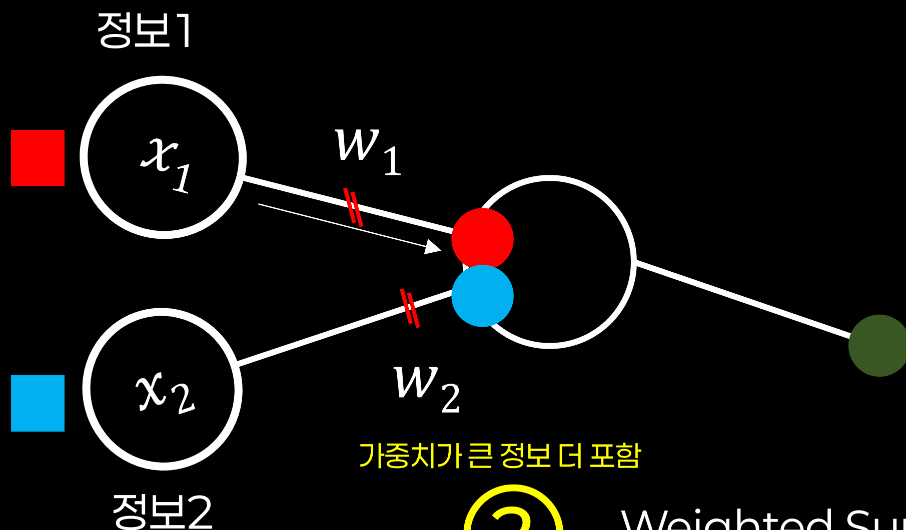


이전 정보 보존

① 정보1 x w_1 한 결과(●)에는 이전 정보1 (x_1)가 보존되어 있다.

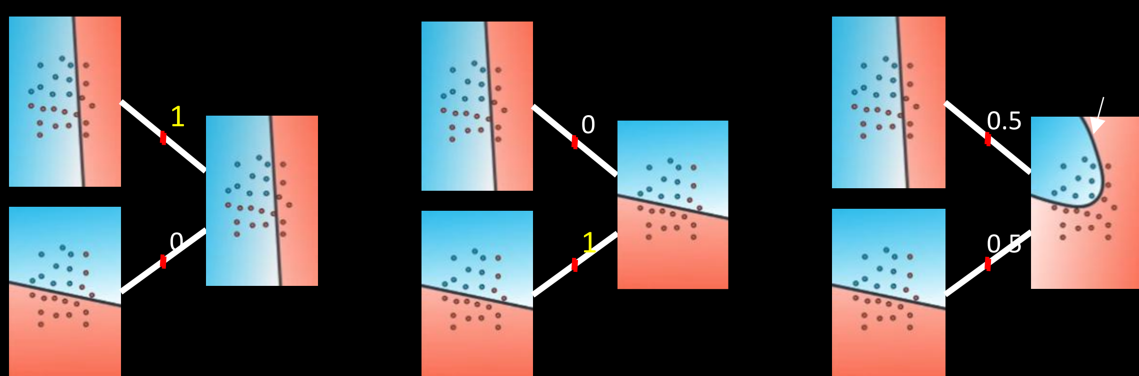
https://www.youtube.com/watch?v=6s69XY025MU&t=5s

벡터의 3가지 성질 (2/3)



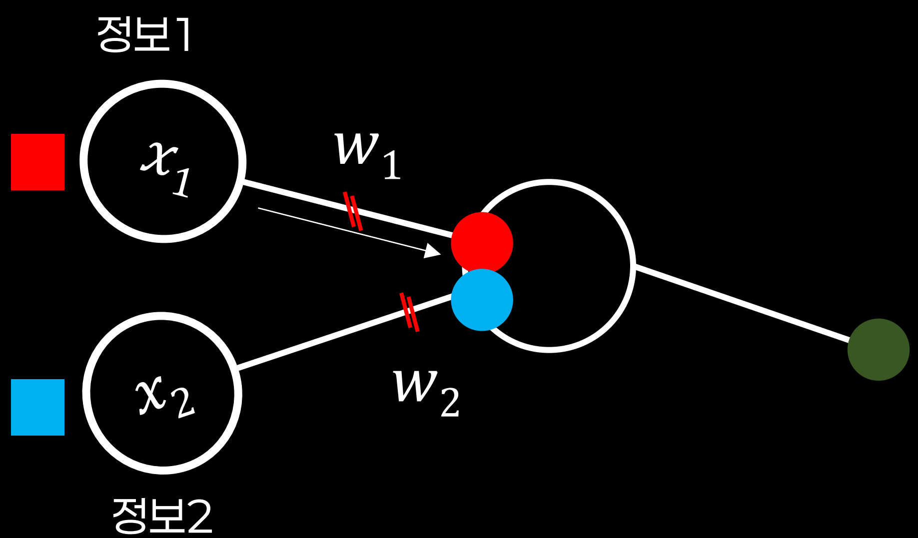
$$(1,1)\begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$$

② Weighted Sum( )에는 가중치가 큰 쪽의 정보를 더 **많이** 갖고 있다.
물론, 작은 쪽 정보는 **적게** 갖고 있다.



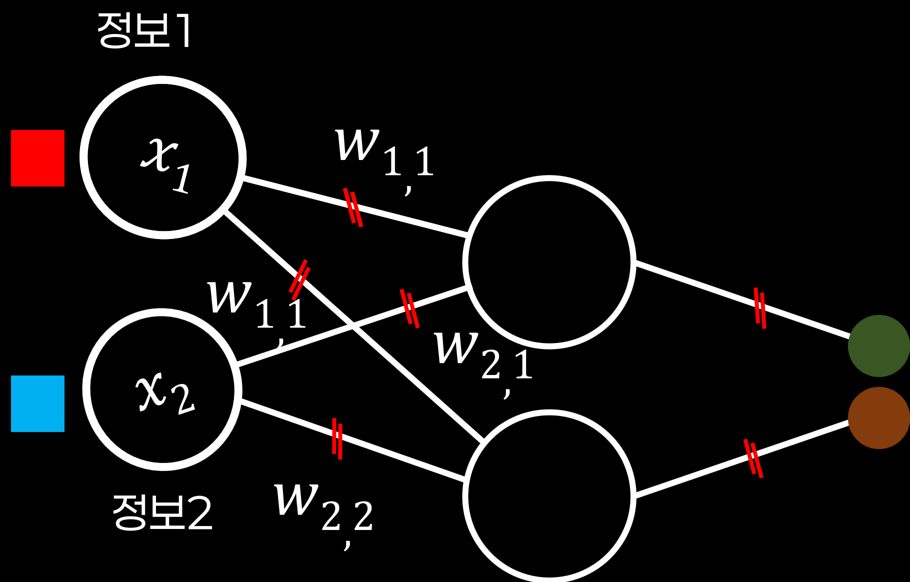
https://www.youtube.com/watch?v=6s69XY025MU&t=5s

벡터의 3가지 성질 (2/3)



$$(1, 1)\begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$$

https://www.youtube.com/watch?v=6s69XY025MU&t=5s

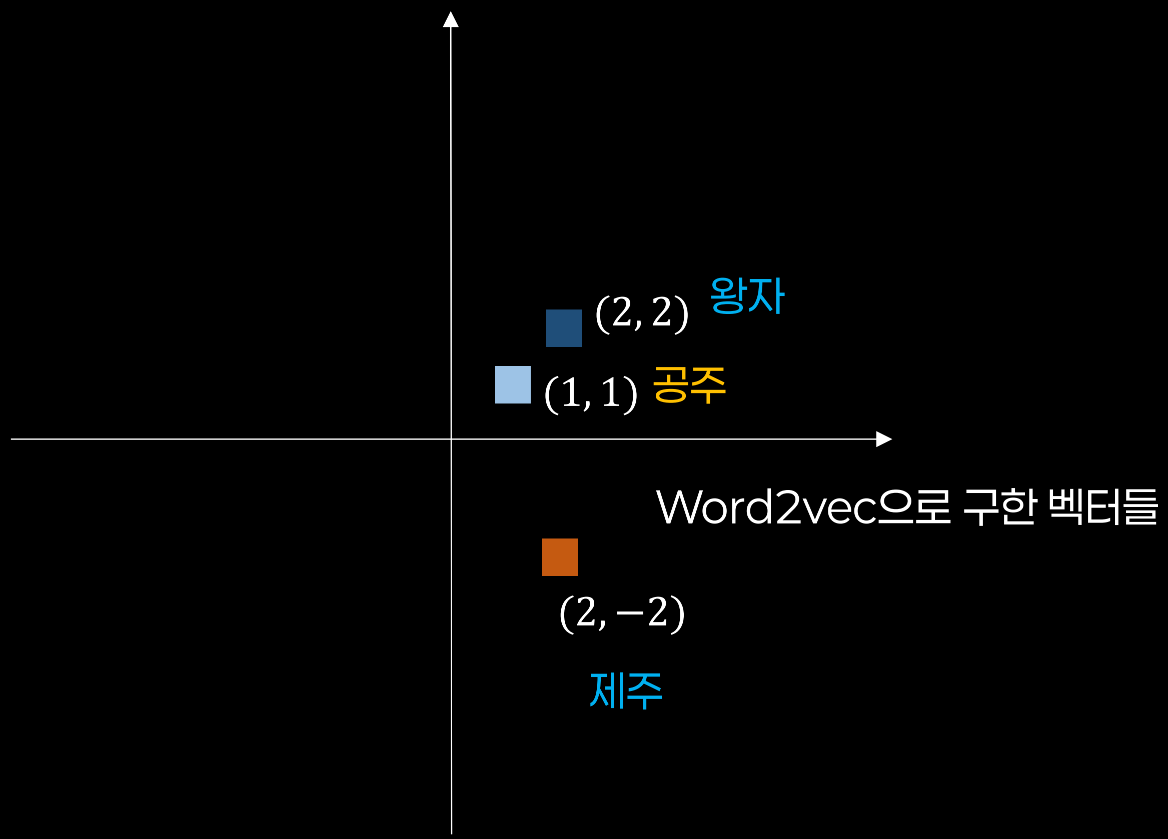


$$(1,1)\begin{pmatrix} w_{1,1} & w_{1,2} \\ w_{2,1} & w_{2,2} \end{pmatrix}$$

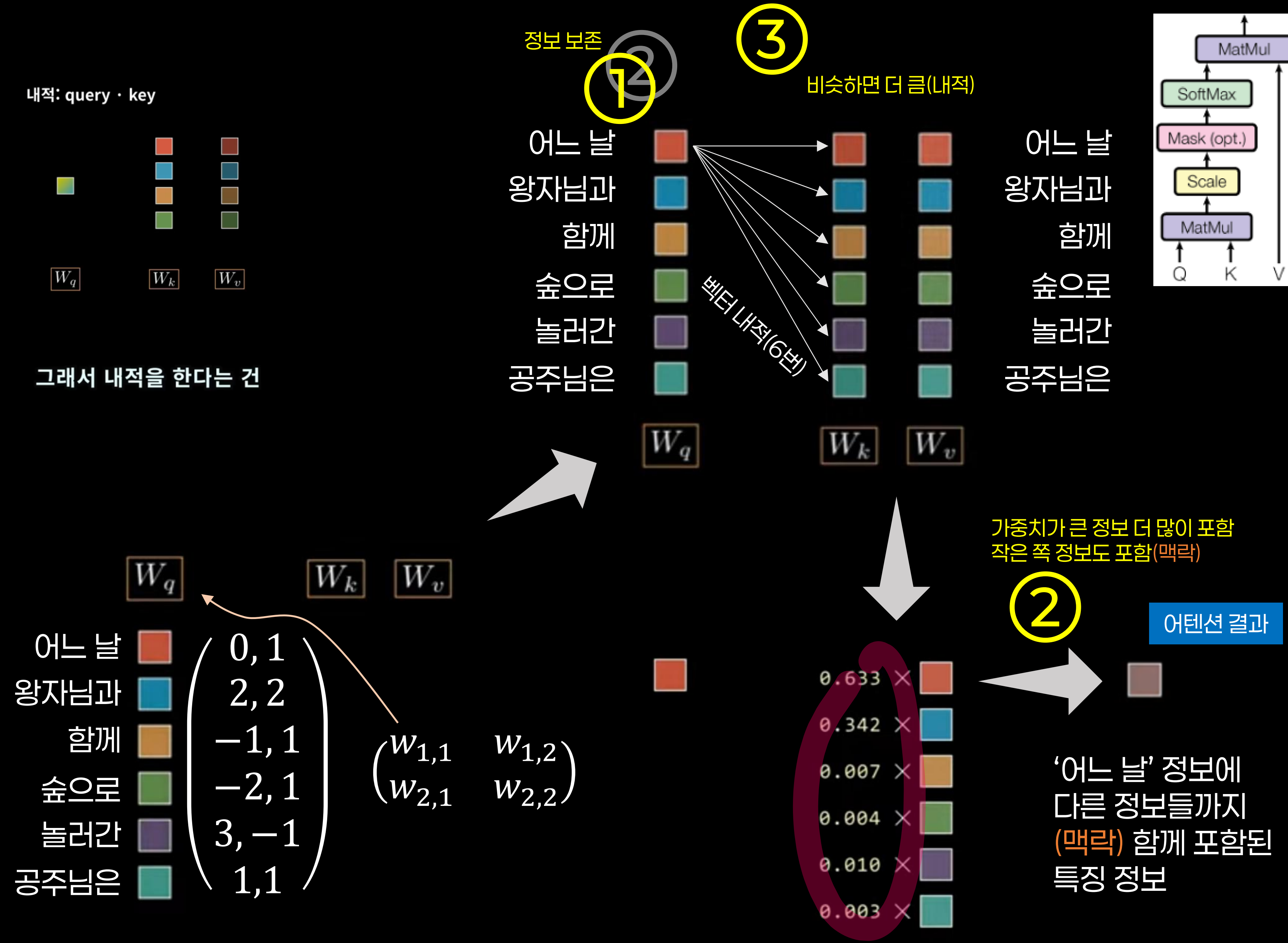
벡터의 3가지 성질 (3/3)

③ 비슷하면 더 큼(내적)

비슷한(관련이 있는) 벡터의 내적 값은 크고, 그렇지 않을 경우 작다.

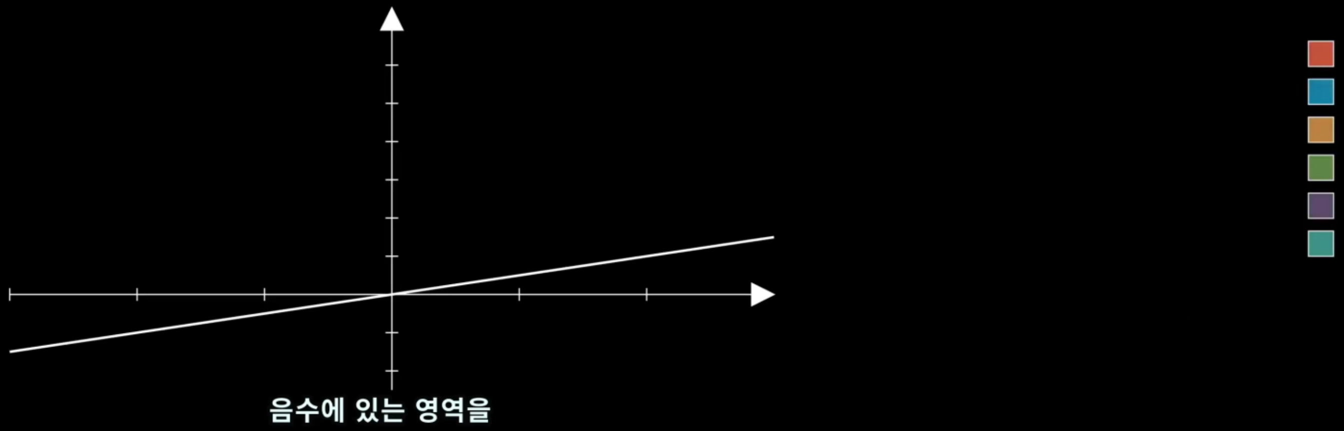


https://www.youtube.com/watch?v=6s69XY025MU&t=5s



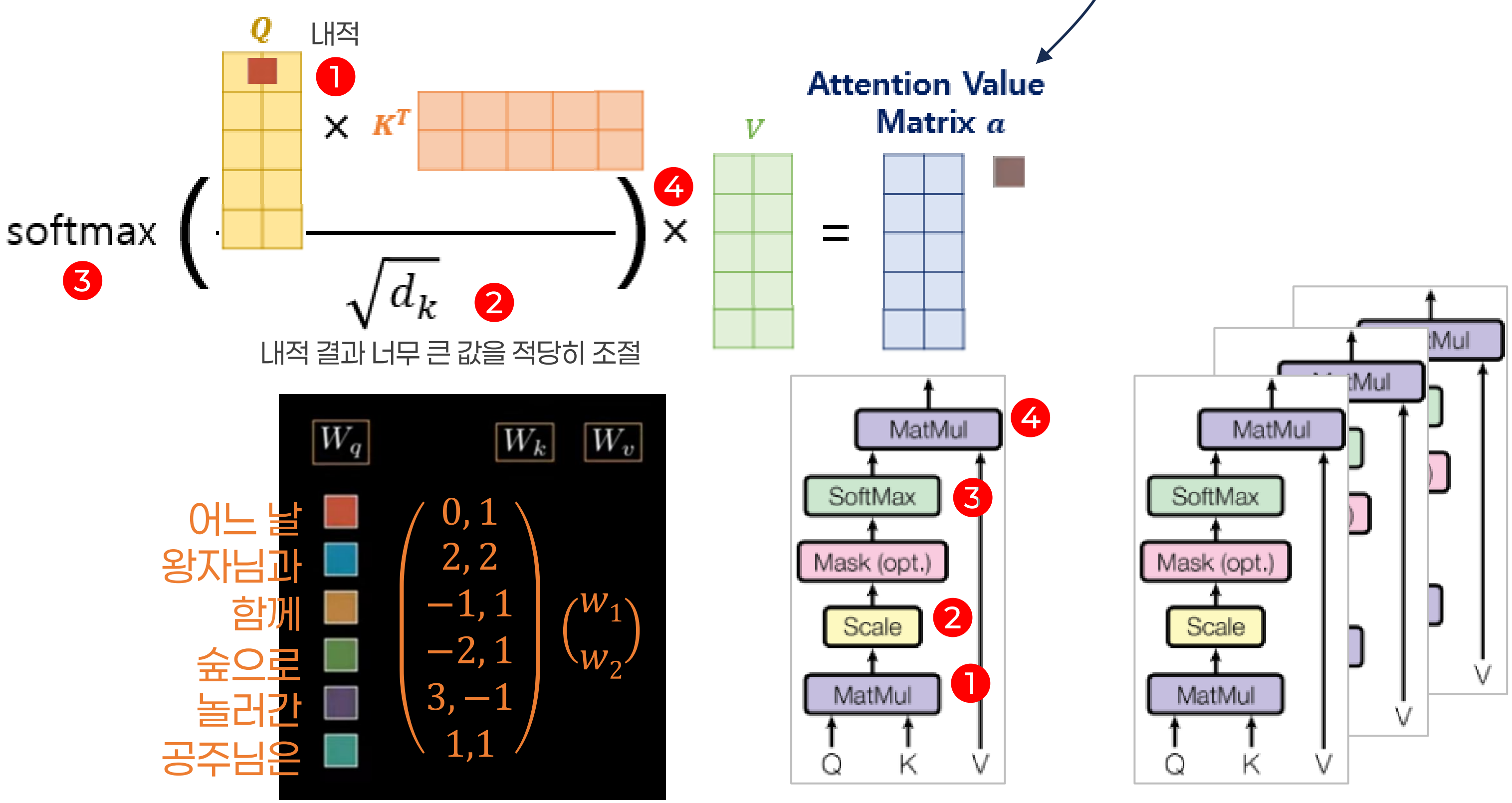
https://www.youtube.com/watch?v=6s69XY025MU&t=5s

벡터의 3가지 성질 (3/3)



$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

이렇게 하여 얻은 어텐션 값 행렬(AVM)을 특징으로 이용하여 정답을 맞춘다. 정답을 잘 맞히도록 시냅스를 조절하는 것이 학습



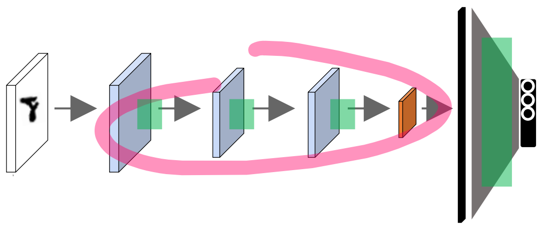
Single-Head Attention
(혹은 바닐라 어텐션)

Multi-Head Attention

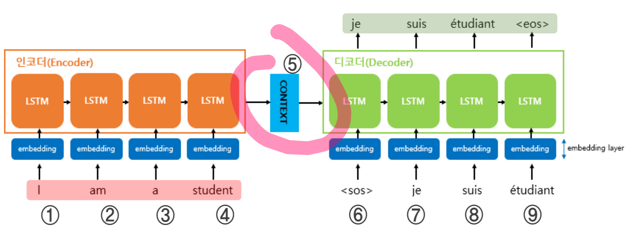
https://codingopera.tistory.com/43

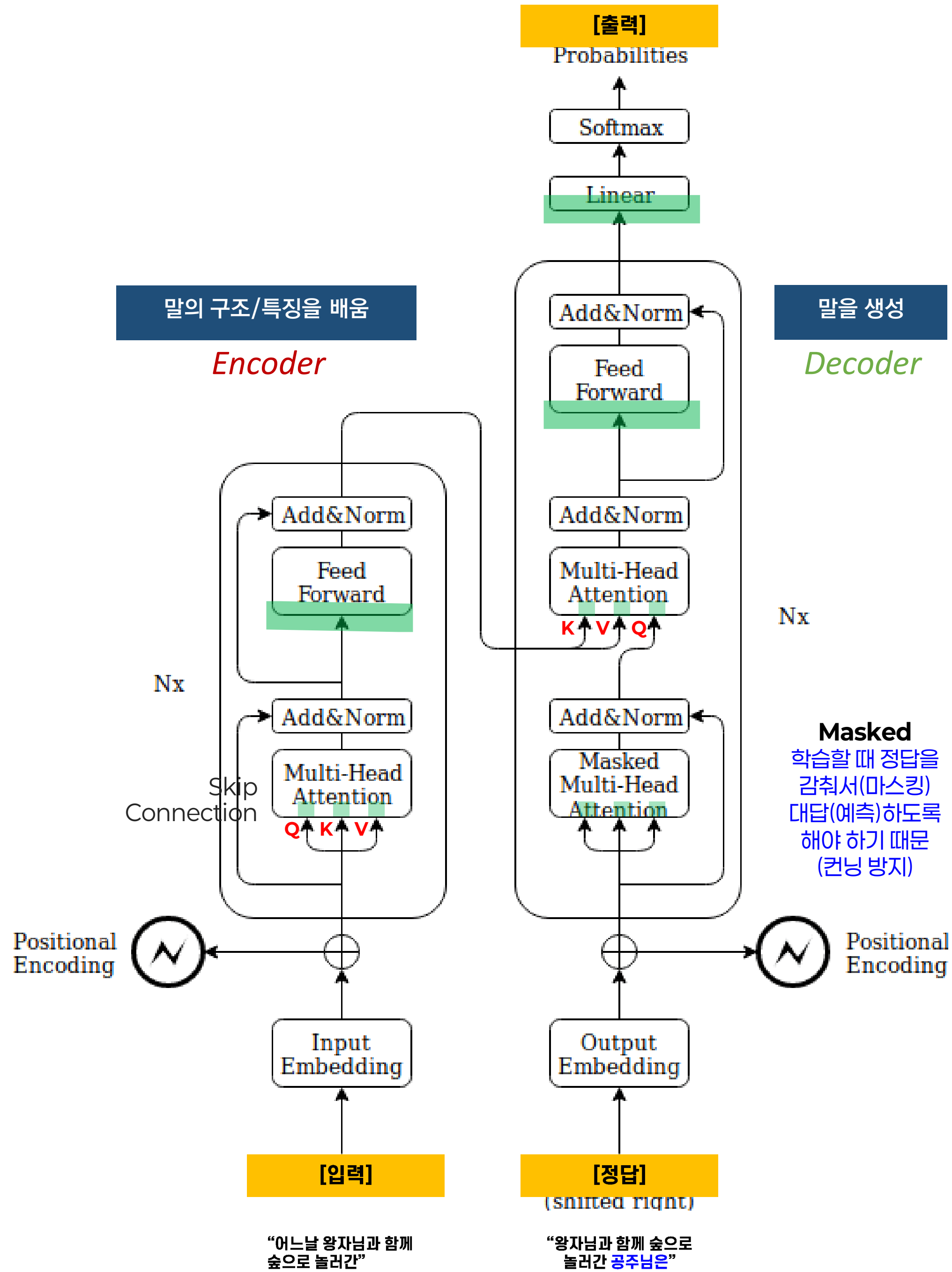
"어느날 왕자님과 함께 숲으로 놀러간 공주님은"

트랜스포머



인코더의
마지막 출력 정보
(압축 정보)





학습시킬 파라미터
(GPT 3.5 - 1,750억 개)

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강력한 AI
예고편이자
변곡점

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인공지능 패러다임과 기업의 흥망

게임 체인저